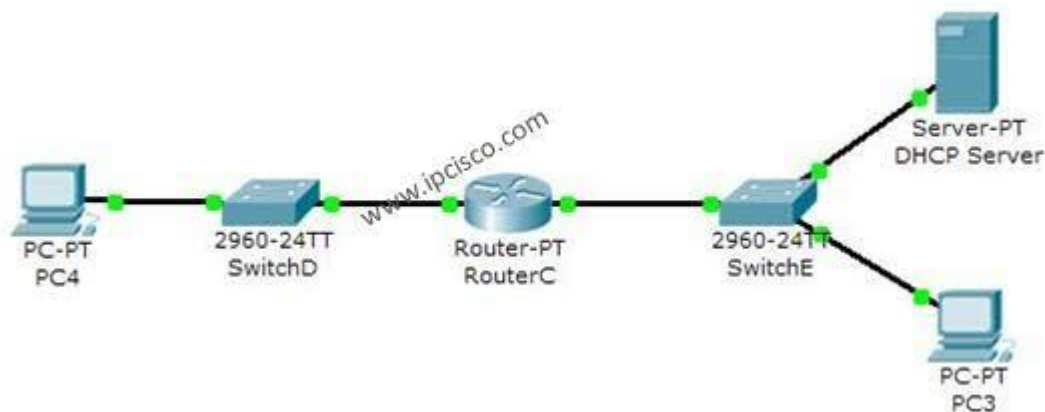


Day 2: DHCP Configuration Theory

Task 1 :- In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

ANS :-

Step 1: Connect the Devices :-



1. Add 1 Switch (e.g., 2960), 1 Server, and 2 PCs (PC0 and PC1) to your workspace.
2. Connect all 3 devices to the switch using Copper Straight-Through cables.

Step 2: Configure the Server IP & DHCP :-

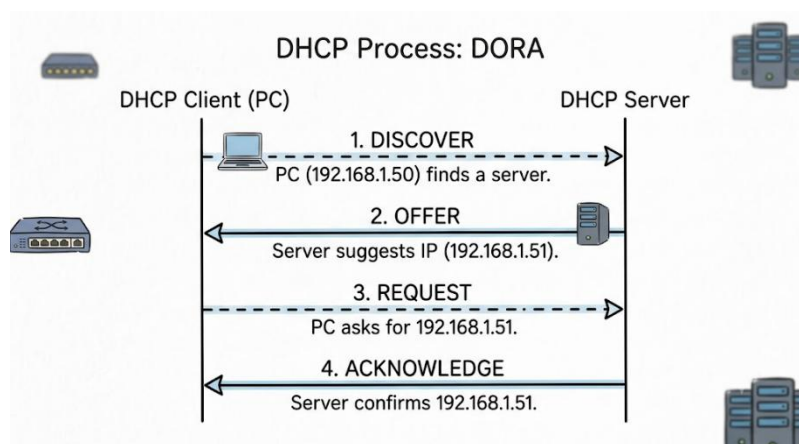
1. Click the Server → Desktop tab → IP Configuration.
 - Select Static and give it an IP: 192.168.1.1 (Subnet Mask: 255.255.255.0).
2. Go to Services tab → DHCP.
 - Turn Service ON.
 - Set Start IP Address to 192.168.1.2.
 - Click Save.

Step 3: Verify Automatic IP Assignment on PCs :-

1. Click PC0 → Desktop tab → IP Configuration.
 - Switch from *Static* to DHCP.
 - Result : It will automatically receive IP 192.168.1.2.
2. Click PC1 → Desktop tab → IP Configuration.
 - Switch from *Static* to DHCP.
 - Result : It will automatically receive IP 192.168.1.3.

Task 2 :- Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.

ANS :-



The DHCP DORA Process : Simple Language :-

1. DISCOVER (Client → All) : Your computer (or phone) just turned on. It shouts, "Is there a DHCP server here? I need an IP address!"
2. OFFER (Server → Client) : A DHCP server replies. It suggests, "I have an IP address available for you! It's 192.168.1.51."
3. REQUEST (Client → Server) : Your computer tells the server, "Okay, that sounds good! I would like to use 192.168.1.51, please."
4. ACKNOWLEDGE (Server → Client) : The server confirms the assignment. It says, "It's yours! Here is the confirmed 192.168.1.51 address and other network details."

Task 3 :- Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

ANS :-

The screenshot displays the Cisco Packet Tracer interface with three main components:

- Router0 CLI:** Shows the configuration of a DHCP pool named 'test' with a range of 192.168.0.0 to 192.168.0.255. The 'show ip dhcp binding' command output shows two active leases: 192.168.0.2 for client 00D0.BC46.E7E9 and 192.168.0.3 for client 0001.6338.7603.
- PC0 Configuration:** Shows the configuration for PC0, which is set to obtain an IP address via DHCP. The IP address is 169.254.231.233 and the subnet mask is 255.255.0.0.
- Network Diagram:** Shows a network topology with two VLANs (VLAN10 - Area1 and VLAN20 - Area2) connected to a central Switch-PT. PC0 is connected to VLAN10 and PC1 to VLAN20. A 2621XM Router0 is connected to the switch via Fa2/1 and Fa0/0, with IP 192.168.0.1 assigned to Fa0/0.

Step-by-Step Setup :-

1. Set Pool Size to 1 : On your DHCP Server, go to Services → DHCP. Set Maximum Number of Users to 1 (or set start IP to 192.168.1.2 and end IP to 192.168.1.2), then click Save.
2. Request IP on PC0 : Open PC0 → Desktop → IP Configuration → Select DHCP. It gets the single available IP (192.168.1.2).
3. Request IP on PC1 : Open PC1 → Desktop → IP Configuration → Select DHCP. It shows "DHCP failed" and gets an APIPA address (169.254.x.x).

Explanation (Why only one PC succeeds) :-

- Reason : A DHCP server hands out IP addresses like ticket numbers from a limited bucket.
- Result : Since you set the pool size to 1, PC0 claimed the only IP available. When PC1 asked for an IP, the pool was completely empty (exhausted), so PC1 failed to get one.

Task 4 :- Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

ANS :-

Step 1: Network Setup :-

1. Place 1 Server (DHCP), 1 Wireless Router (or Access Point), 1 Laptop, and 1 PC (acting as Smartphone).
2. Connect the Server to the Wireless Router with a straight-through cable.
3. Set Server IP statically to 192.168.1.1 and enable DHCP Service (Start IP: 192.168.1.100).

Step 2 : Assigned Ips :-

- Laptop : 192.168.1.100 (Assigned automatically via Wi-Fi)
- Smartphone (PC) : 192.168.1.101 (Assigned automatically via Wi-Fi)

Why DHCP is useful here :-

- No Manual Work : Customers constantly enter and leave the restaurant; DHCP gives them instant internet access without staff typing IP settings manually on every phone.
- IP Reuse : When a customer leaves, DHCP reclaims their IP address and gives it to the next customer.

